```python
# Online Python compiler (interpreter)
# Write and run Python online using this editor.

import math

c = input("Enter ciphertext: ").upper()

for a in range(1, 26):
    if math.gcd(a, 26) == 1:
        for b in range(26):
            p = ""
            inv = pow(a, -1, 26)

            for ch in c:
                if ch.isalpha():
                    x = ord(ch) - 65
                    p += chr(((inv * (x - b)) % 26) + 65)
                else:
                    p += ch

            if p:
                print("a =", a, "b =", b, "Plaintext =", p)
```

```
Enter ciphertext: b
a = 1 b = 0 Plaintext = B
a = 1 b = 1 Plaintext = A
a = 1 b = 2 Plaintext = Z
a = 1 b = 3 Plaintext = Y
a = 1 b = 4 Plaintext = X
a = 1 b = 5 Plaintext = W
a = 1 b = 6 Plaintext = V
a = 1 b = 7 Plaintext = U
a = 1 b = 8 Plaintext = T
a = 1 b = 9 Plaintext = S
a = 1 b = 10 Plaintext = R
a = 1 b = 11 Plaintext = Q
a = 1 b = 12 Plaintext = P
a = 1 b = 13 Plaintext = O
a = 1 b = 14 Plaintext = N
a = 1 b = 15 Plaintext = M
a = 1 b = 16 Plaintext = L
a = 1 b = 17 Plaintext = K
a = 1 b = 18 Plaintext = J
a = 1 b = 19 Plaintext = I
a = 1 b = 20 Plaintext = H
```

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            for ch in c:
                if ch.isalpha():
                    x = ord(ch) - 65
                    p += chr(((inv * (x - b)) % 26) + 65)
                else:
                    p += ch

            if p:
                print("a =", a, "b =", b, "Plainte p)
```

Search

```
a = 25 b = 5 Plaintext = E
a = 25 b = 6 Plaintext = F
a = 25 b = 7 Plaintext = G
a = 25 b = 8 Plaintext = H
a = 25 b = 9 Plaintext = I
a = 25 b = 10 Plaintext = J
a = 25 b = 11 Plaintext = K
a = 25 b = 12 Plaintext = L
a = 25 b = 13 Plaintext = M
a = 25 b = 14 Plaintext = N
a = 25 b = 15 Plaintext = O
a = 25 b = 16 Plaintext = P
a = 25 b = 17 Plaintext = Q
a = 25 b = 18 Plaintext = R
a = 25 b = 19 Plaintext = S
a = 25 b = 20 Plaintext = T
a = 25 b = 21 Plaintext = U
a = 25 b = 22 Plaintext = V
a = 25 b = 23 Plaintext = W
a = 25 b = 24 Plaintext = X
a = 25 b = 25 Plaintext = Y
```